Assignment – 5

Q1. Create a tuple with different data and create an empty tuple using built-in function.

```
tup = eval(input("Enter a tuple with different elements: "))
print("Tuple with different elements:", tup)
```

Output:

```
Enter a tuple with different elements: 1, 0.5, "Hello"
Tuple with different elements: (1, 0.5, 'Hello')
```

Q2. Print the 3rd element from beginning and 4th element from end of a tuple.

```
tup = eval(input("Enter the tuple of elements: "))

if len(tup) <= 3:
    print("There must be more than 3 elements")
else:
    print("The third element from the beginning:", tup[2])
    print("The fourth element from the end:", tup[-4])
```

Output:

```
Enter the tuple of elements: 1, 2, 3, 4, 5
The third element from the beginning: 3
The fourth element from the end: 2
```

Q3. Count the number of items present in a tuple and then add two new items into it (one item using concatenation operator and another item using built-in function).

```
tup = eval(input("Enter the tuple of elements: "))
print("There are", len(tup), "elements in the tuple")

tup += (eval(input("Enter a element to add: ")),)
lst = list(tup)
lst.append(eval(input("Enter another element to add: ")))
tup = tuple(lst)

print(tup)
```

Output:

```
Enter the tuple of elements: 1, 2, 3, 4
There are 4 elements in the tuple
Enter a element to add:  5
Enter another element to add:  6
(1, 2, 3, 4, 5, 6)
```

Q4. Find the number of occurrences for any repeated item present in a tuple.

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```
tup = eval(input("Enter the tuple of elements: "))
seen = []

for element in tup:
    if element not in seen:
        print(element, ":", tup.count(element))
        seen.append(element)
```

Output:

```
Enter the tuple of elements: 2, 5, 0, 3, 8, 6, 6, 0
2 : 1
5 : 1
0 : 2
3 : 1
8 : 1
6 : 2
```

Q5. Display the elements present in a tuple which are at the even position.

```
tup = eval(input("Enter the tuple of elements: "))
print("Elements present at even positions are:", tup[::2])
```

Output:

```
Enter the tuple of elements: 2, 5, 0, 3, 8, 6, 6, 0
Elements present at even positions are: (2, 0, 8, 6)
```

Q6. Remove the elements of a tuple which are present in the odd index position.

```
tup = eval(input("Enter the tuple of elements: "))
tup = tup[::2]
print("After removing elements at odd index:", tup)
```

Output:

```
Enter the tuple of elements: 2, 5, 0, 3, 8, 6, 6, 0
After removing elements at odd index: (2, 0, 8, 6)
```

Q7. Remove the odd number elements from a tuple.

```
tup = eval(input("Enter the tuple of elements: "))
tup = tuple(element for element in tup if element % 2 == 0)
print("After removing odd number elements:", tup)
```

Output:

```
Enter the tuple of elements: 2, 5, 0, 3, 8, 6, 6, 0
After removing odd number elements: (2, 0, 8, 6, 6, 0)
```

Q8. Reverse the elements of a tuple by using and without using a built-in function.

```
tup = eval(input("Enter the tuple of elements: "))
print("Reversed using built-in function:", tuple(reversed(tup)))
print("Reversed without using built-in function:", tup[::-1])
```

Output:

```
Enter the tuple of elements: 2, 5, 0, 3, 8, 6, 6, 0
Reversed using built-in function: (0, 6, 6, 8, 3, 0, 5, 2)
Reversed without using built-in function: (0, 6, 6, 8, 3, 0, 5, 2)
```

Q9. Sort the elements present in a tuple in descending order.

```
tup = eval(input("Enter the tuple of elements: "))
print("Elements sorted in descending order:", tuple(sorted(tup, reverse=True)))
```

Output:

```
Enter the tuple of elements: 2, 5, 0, 3, 8, 6, 6, 0
Elements sorted in descending order: (8, 6, 6, 5, 3, 2, 0, 0)
```

Q10. Take a tuple of mixed data types and find the square of each element.

```
tup = eval(input("Enter the tuple of elements: "))
print("Square of each element is:", tuple(element ** 2 for element in tup))
```

Output:

```
Enter the tuple of elements: 2, 5, 0, 3, 8, 6, 6, 0
Square of each element is: (4, 25, 0, 9, 64, 36, 36, 0)
```

Q11. Given few numbers in a tuple, return 'True' if the first and last numbers are same.

```
tup = eval(input("Enter the tuple of elements: "))
print(tup and tup[0] == tup[-1])
```

Output:

```
Enter the tuple of elements: 2, 5, 0, 3, 8, 6, 6, 0
False
```

Q12. Print only those elements present in a tuple which are divisible by 5.

```
tup = eval(input("Enter the tuple of elements: "))
print("Elements divisible by 5 are:", tuple(element for element in tup if element % 5 == 0))
```

Output:

```
Enter the tuple of elements: 2, 5, 0, 3, 8, 6, 6, 0
Elements divisible by 5 are: (5, 0, 0)
```

Q13. Given a tuple of numbers, create another tuple such that it contains only the even numbers of the first tuple.

```
tup = eval(input("Enter the tuple of elements: "))
new_tup = tuple(element for element in tup if element % 2 == 0)
print("Tuple with only even numbers are:", new_tup)
```

Output:

```
Enter the tuple of elements: 2, 5, 0, 3, 8, 6, 6, 0
Tuple with only even numbers are: (2, 0, 8, 6, 6, 0)
```

Q14. Display the largest number of a tuple without using the built-in function max().The user need to input the values in tuple from the keyboard.

```
tup = eval(input("Enter the tuple of elements: "))
res = tup[0]

for element in tup[1:]:
    if element > res:
        res = element

print("The largest element is:", res)
```

Output:

```
Enter the tuple of elements: 2, 5, 0, 3, 8, 6, 6, 0
The largest element is: 8
```

Q15. Replace the last value of the tuple with 100.
Sample list: [(1, 2, 3), (4, 5, 6), (7, 8, 9)]
Output: [(1, 2, 100), (4, 5, 100), (7, 8, 100)]

```
lst = list(eval(input("Enter the list of tuples: ")))

for i in range(len(lst)):
    lst[i] = lst[i][:-1] + (100,)

print("After replacing last element with 100:", lst)
```

Output:

```
Enter the list of tuples: [(1, 2, 3), (4, 5, 6), (7, 8, 9)]
After replacing last element with 100: [(1, 2, 100), (4, 5, 100), (7, 8, 100)]
```

Q16. Sort a tuple by its float element.
Sample data: [('item1', '12.20'), ('item2', '15.10'), ('item3', '24.5')]
Output: [('item3', '24.5'), ('item2', '15.10'), ('item1', '12.20')]

```
lst = list(eval(input("Enter the list of tuples: ")))
lst.sort(key=lambda element : element[1], reverse=True)
print("After sorting:", lst)
```

Output:

```
Enter the list of tuples: [('item1', '12.20'), ('item2', '15.10'), ('item3', '24.5')]
After sorting: [('item3', '24.5'), ('item2', '15.10'), ('item1', '12.20')]
```

Q17. Count the number of elements in a tuple until an element is a list.

```
tup = eval(input("Enter the tuple of elements: "))
res = 0

for element in tup:
    if type(element) == list:
        break

    res += 1

print("Count is:", res)
```

Output:

```
Enter the tuple of elements: 2, 5, 0, 3, (8, 6, 6, 0)
Count is: 5
```

Q18. Swap the two tuples:

Sample: tuple1 = (11, 22) tuple2 = (99, 88)

Output: tuple1 = (99, 88) tuple2 = (11, 22)

```
tup1 = eval(input("Enter the tuple of elements: "))
tup2 = eval(input("Enter the tuple of elements: "))

if tup1 < tup2:
    tup1, tup2 = tup2, tup1

print("After sorting:", tup1, tup2)
```

Output:

```
Enter the tuple of elements:  2, 5, 0, 3
Enter the tuple of elements:  8, 6, 6, 0
After sorting: (8, 6, 6, 0) (2, 5, 0, 3)
```

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